

DASHBOARD DEVELOPMENT FOR REAL-TIME WEATHER FORECASTING AND ANALYSIS USING POWER BI AND API INTEGRATION

In partial fulfilment for the award of

Bachelor of Technology in

Artificial Intelligence in Data Science

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CERTIFICATE

This is to certify that, the mini-project entitled “**DASHBOARD DEVELOPMENT FOR REAL-TIME WEATHER FORECASTING AND ANALYSIS USING POWER BI AND API INTEGRATION**”, which has been submitted herewith in the partial fulfilment for the award of the ‘Bachelor of Technology’ in ‘Artificial Intelligence in Data Science’ of Dr. Babasaheb Ambedkar University, Chh. Sambhajinagar (M.S.). This is the result of the original work and contribution for Project by Vijay Nagargoje and Shubham Patange under my supervision and guidance.

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ABSTRACT

Accurate weather monitoring and forecasting are critical for agriculture, urban planning, and daily decision-making. This project, "Dashboard Development for Real-Time Weather Forecasting and Analysis using Power BI and API Integration," aims to develop an automated, interactive meteorological tracking system. Instead of relying on static historical datasets, this project utilizes a custom mini-ETL (Extract, Transform, Load) pipeline to pull live 7-day forecast and current weather data from a RESTful Weather API.

The data encompasses multiple key cities in Maharashtra, including Nanded, Aurangabad, Jalna, and Mumbai. The extracted data was modeled into a comprehensive semantic structure and analyzed using Microsoft Power BI. The resulting interactive dashboard provides real-time insights across several critical dimensions: temperature forecasting, air quality indices (AQI), atmospheric pressure, humidity, and wind speed.

The findings demonstrate the capability to dynamically track regional climate variations, highlighting peak temperatures approaching 38.5°C and providing immediate visibility into localized air quality conditions. This dynamic dashboard serves as a functional tool for accessible, data-driven weather analysis, demonstrating the practical application of API-driven data engineering.

Chapter 1: Introduction

Weather conditions significantly impact various sectors, including agriculture, urban planning, disaster management, and daily decision-making. With the advent of modern RESTful APIs, access to real-time meteorological data has become more accessible, yet making sense of this raw, nested data requires robust analytical tools. This project develops an interactive analytics dashboard using Microsoft Power BI based on dynamic data fetched via a Weather API. It transforms live weather metrics and forecasts into clear visual insights, helping users monitor regional climate variations efficiently.

1.1 Objective of the Project

The primary objectives of this project are as follows:

- Develop an automated mini-ETL pipeline to extract 7-day weather forecast and current condition data using a RESTful Weather API.
- Create a robust semantic data model in Microsoft Power BI to integrate and process the multi-city data.
- Design an interactive dashboard to visualize current and forecasted temperatures for major cities in Maharashtra, including Nanded, Aurangabad, Jalna, and Mumbai.
- Track and display supplementary meteorological metrics including Air Quality Index (AQI), humidity, wind speed, and atmospheric pressure.

1.2 Importance of Data Visualization

Data visualization uses charts and graphs to show data clearly, helping identify patterns and trends, especially for analyzing traffic safety.

- **Pattern Recognition:** Visualization simplifies large weather datasets and helps identify patterns like temperature spikes, seasonal shifts, and geographic climate differences.
- **Decision Support:** Visual dashboards enable quick understanding of complex meteorological data, allowing users to make informed decisions without requiring technical expertise.
- **Communication:** Charts and graphs communicate insights more effectively than raw API JSON outputs, improving clarity in reports and discussions.
- **Real-time Monitoring:** When connected to live API data, dashboards help track environmental conditions (like poor air quality or severe temperatures) in near real-time.

1.3 Introduction to Power BI

Microsoft Power BI is a cloud-based business analytics tool that enables users to create interactive reports and dashboards easily. Launched in 2014, it has become one of the most widely used BI platforms and is consistently recognized as a Leader in the Gartner Magic Quadrant.

Key Components:

- Power BI Desktop: Used for data modeling and dashboard creation

- Power BI Service: Used to publish and share reports online
- Power BI Mobile: Provides access to dashboards on mobile devices
- Power Query: Used for data cleaning and transformation (ETL)
- DAX (Data Analysis Expressions): Used to create calculations, measures, and KPIs

Key Features:

- Connects to 100+ data sources
- Handles large datasets efficiently using in-memory processing
- Creates visually rich and interactive reports

1.4 Overview of Dataset

Unlike static CSV files, this project utilizes dynamic data sourced directly via a Weather API. The dataset represents a live, 7-day rolling window of meteorological data for distinct cities.

Attribute	Description
Location	City Name (e.g., Nanded, Mumbai, Aurangabad, Jalna)
Temperature	Forecasted daily high and low values in Celsius (e.g., peaking at 38.5°C)
Atmospheric Conditions	Humidity percentage, Pressure readings, and Wind Speed in Kph
Environmental Health	Air Quality Index metrics including PM10 and O3
Temporal Data	Sunrise and Sunset times (e.g., Sunrise 05:47 AM, Sunset 06:45 PM), alongside multi-day forecast dates

Chapter 2: Project Methodology

This chapter describes the systematic methodology employed to construct the automated data pipeline and the analytical Power BI dashboard. The methodology follows a modern data engineering pipeline encompassing API data acquisition, JSON transformation, semantic data modeling, and visualization development..

2.1 Dataset Description

2.1.1 Source and Availability

Unlike traditional projects that rely on static CSV files, this project dynamically acquires data using a RESTful Weather API. The API serves as a live data source, ensuring the dashboard reflects real-time meteorological conditions and continuous 7-day forecasting rather than historical, isolated events.

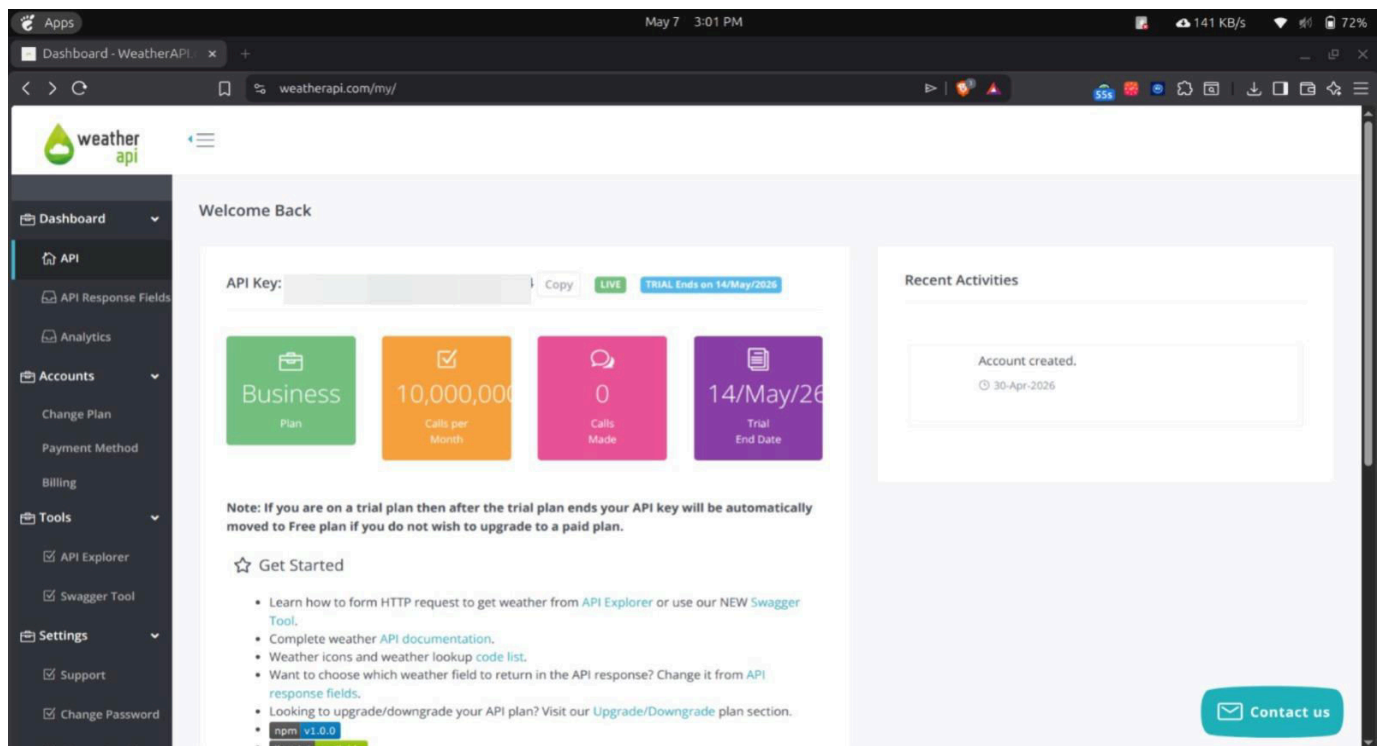


Figure 2.1 : Weather API Interface

2.1.2 Dataset Size and Structure

The dataset is highly dynamic. For this project, the API was configured to retrieve a rolling 7-day data window for six distinct cities in Maharashtra (including Aurangabad, Jalna, Nagpur, Nanded, Mumbai, and Pune). The data is ingested in a nested JSON (JavaScript Object Notation) format, which consists of hierarchical records and lists containing hourly and daily weather metrics.

2.1.3 Key Attributes Used for Analysis

To ensure an efficient and focused semantic model, only the most relevant environmental attributes were extracted from the API payload:

- **Temporal:** Forecast Date, Sunrise Time, Sunset Time.
- **Temperature:** Current Temperature, High, and Low values (in Celsius).
- **Atmospherics:** Humidity (%), Atmospheric Pressure, Wind Speed (Kph).
- **Air Quality:** Air Quality Index (AQI), PM10, PM2.5, SO2, CO, and NO2 levels.

2.2 Data Cleaning and Preprocessing Steps

Data preprocessing is an essential step to ensure data quality and accurate analysis. Because the data originates from an API, the preprocessing was handled entirely within Microsoft Power Query.

2.2.1 API Integration and JSON Parsing

Web queries were configured in Power BI to securely fetch data from the API endpoints for all six cities. Since the raw response is in JSON format, Power Query was used to drill down into the payload. Nested lists containing daily forecast arrays and current weather records were expanded into a flat, tabular format suitable for business intelligence modeling. Once the raw JSON responses were initially parsed and expanded, the data for all six cities was appended and combined into a single, comprehensive dataset named Master_Table.

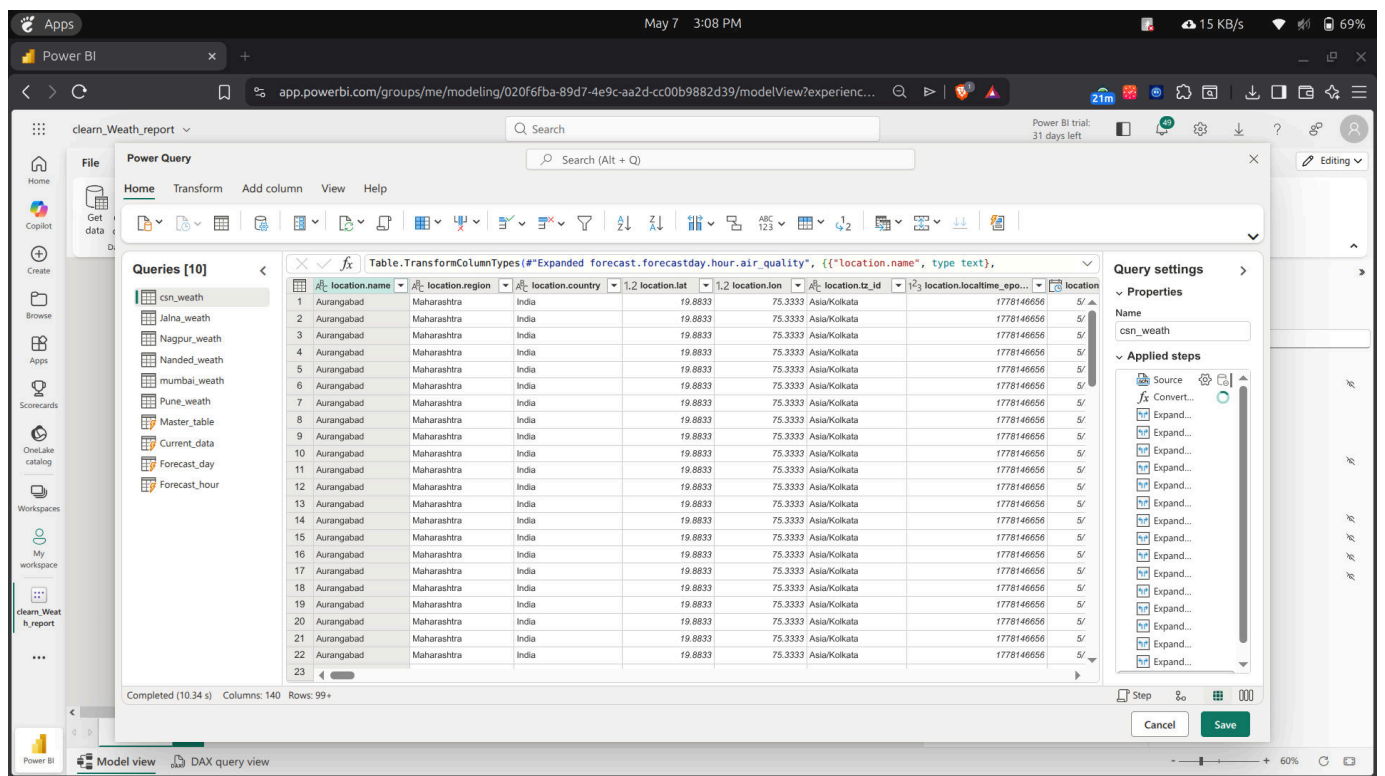


Figure 2.2 : Csn / Aurangabad Data Json Parsing using Power Query

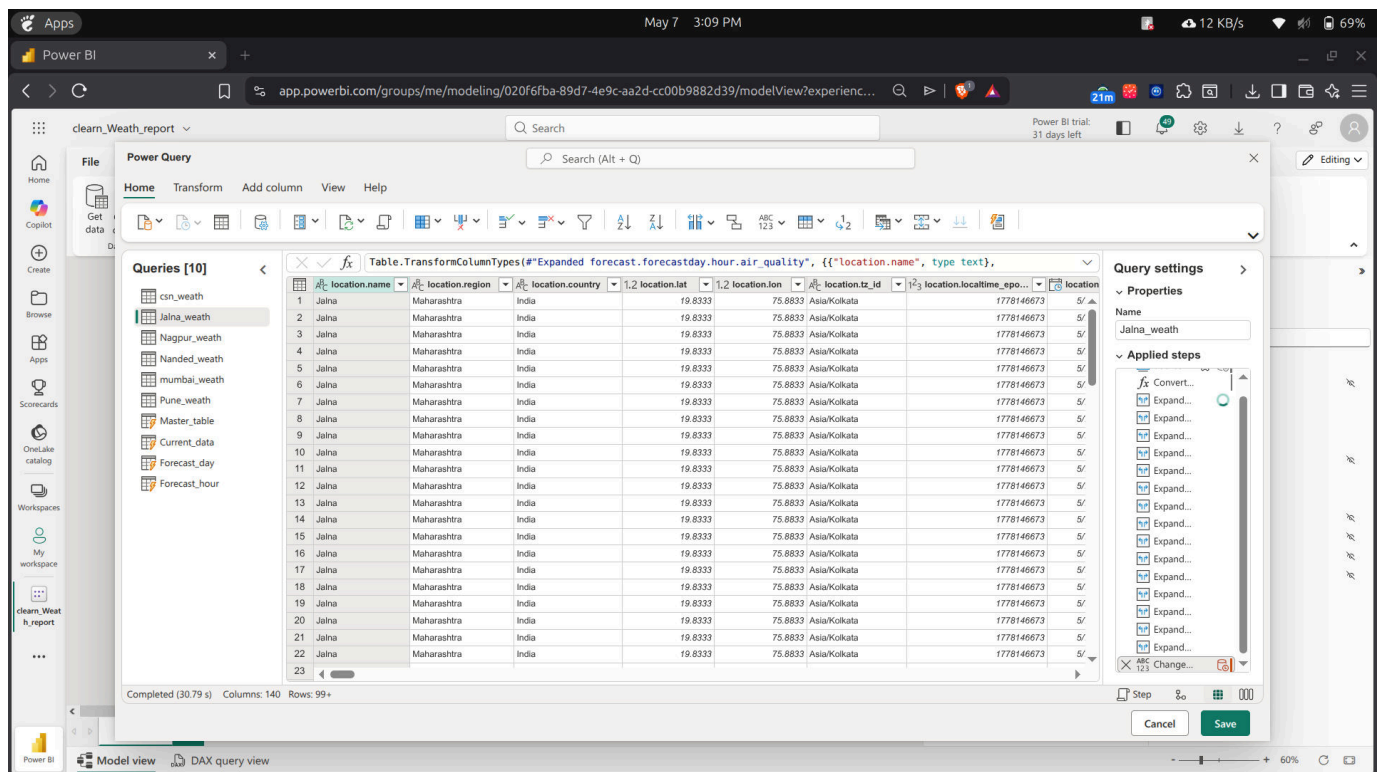


Figure 2.3 : Jalna Data Json Parsing using Power Query

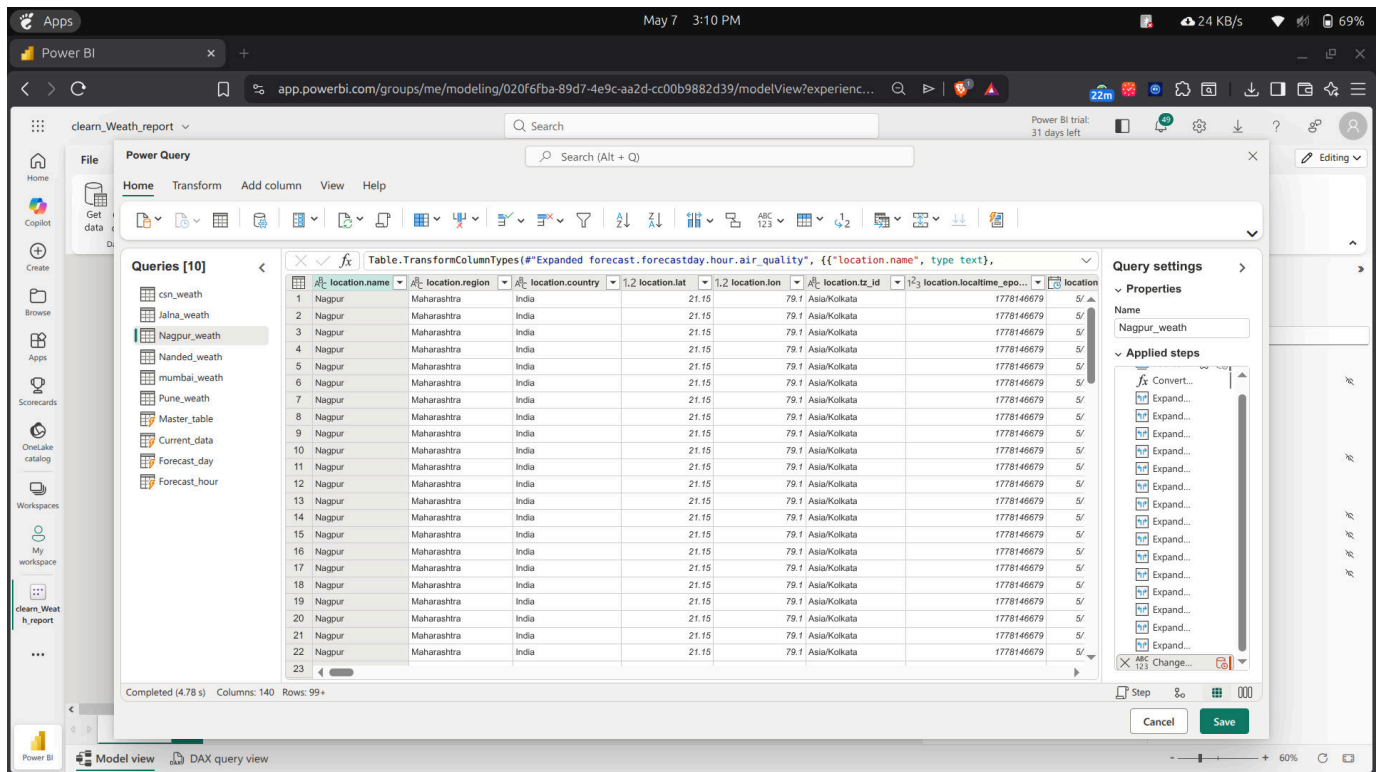


Figure 2.4 : Nagpur Data Json Parsing using Power Query

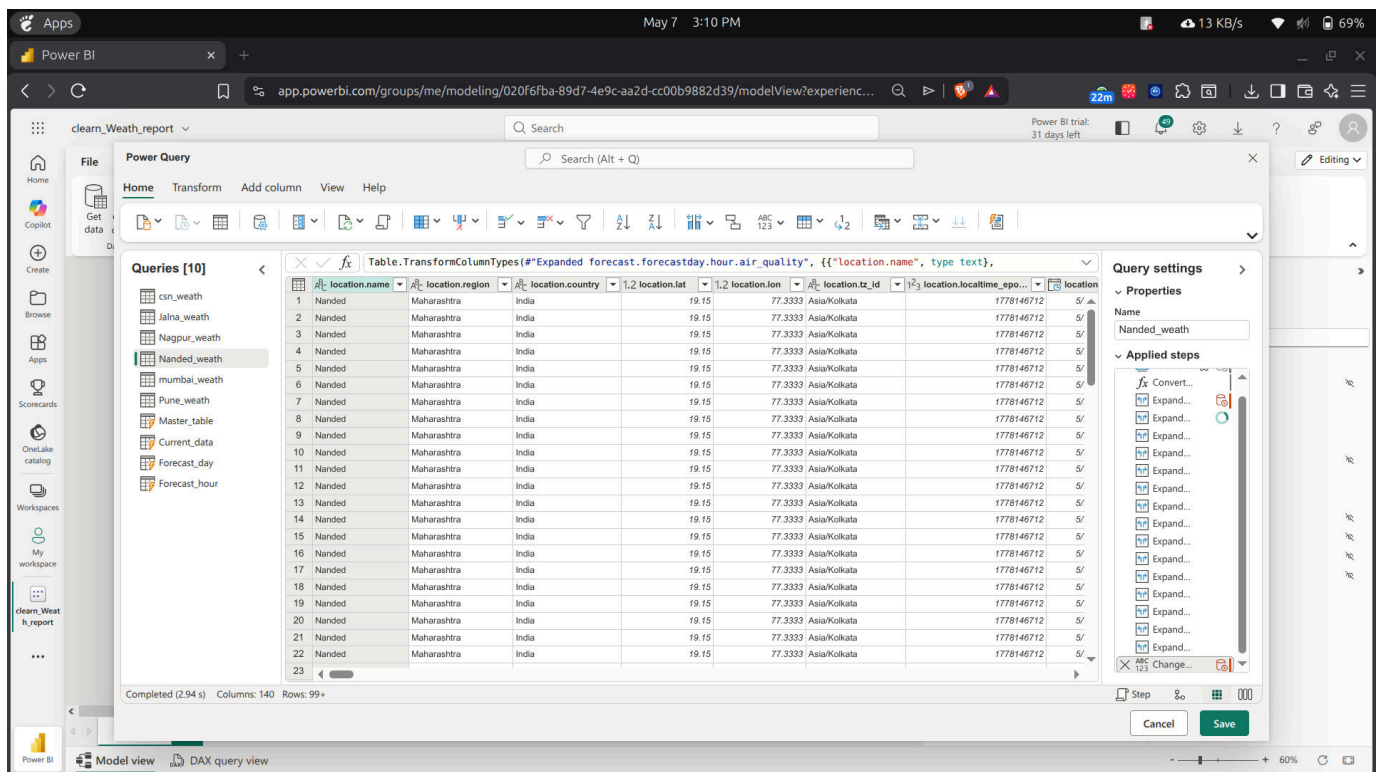


Figure 2.5 : Nanded Data Json Parsing using Power Query

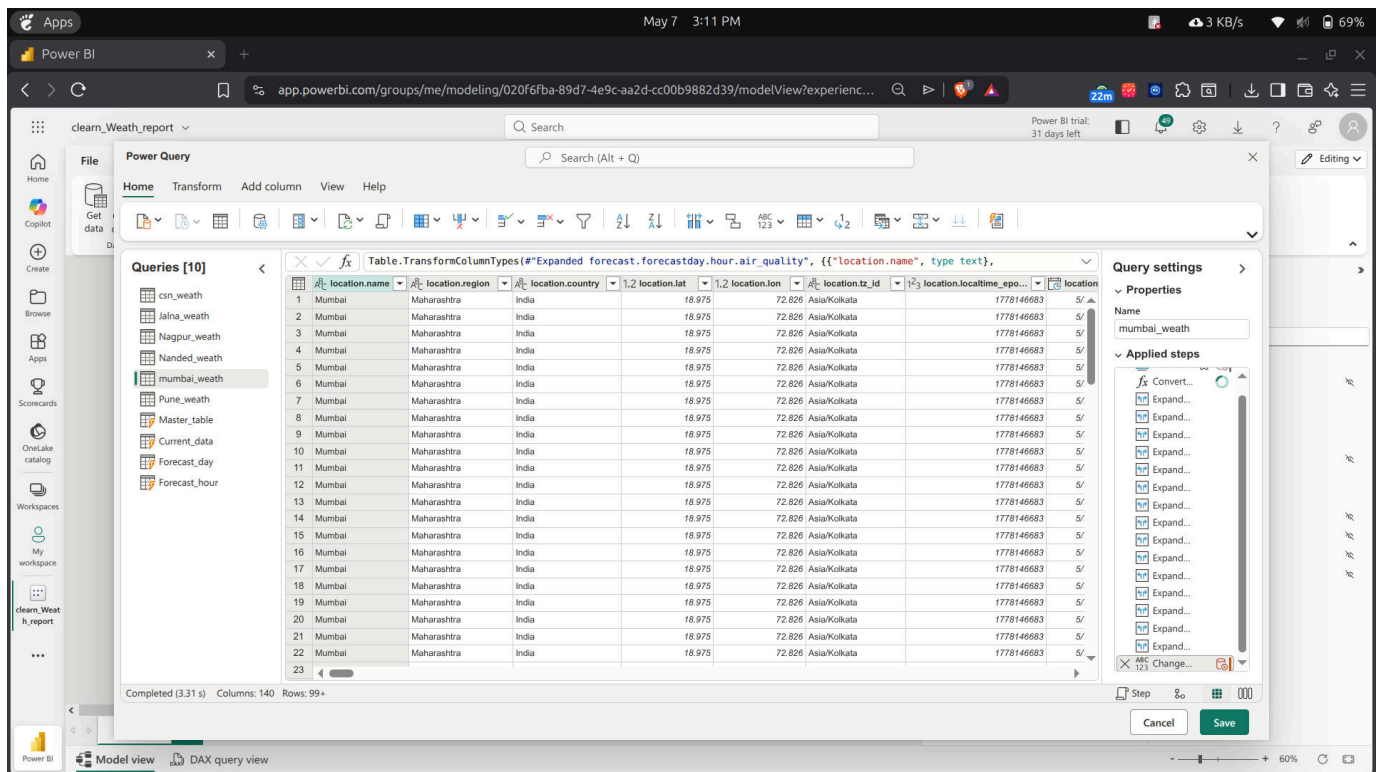


Figure 2.6 : Mumbai Data Json Parsing using Power Query

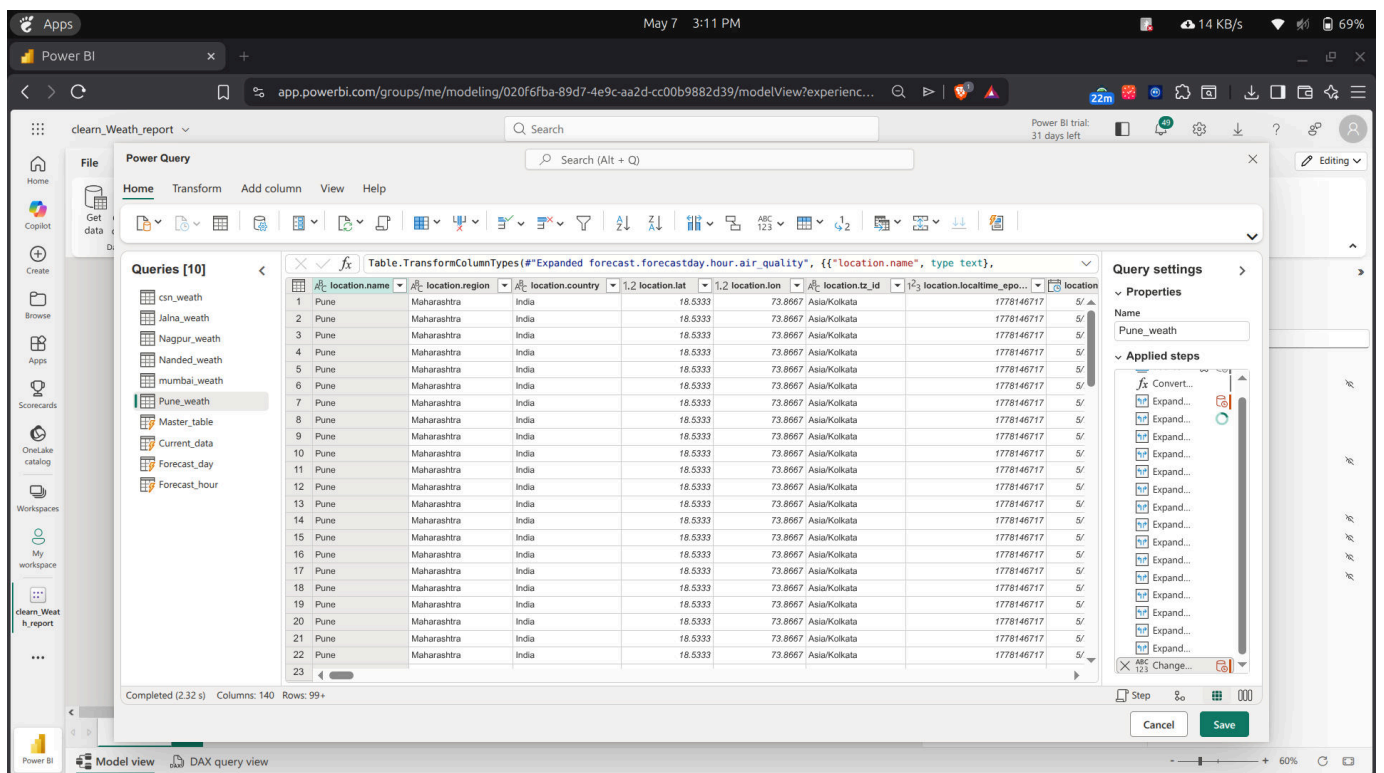


Figure 2.7 : Pune Data Json Parsing using Power Query

2.2.2 Table Derivation and Schema Structuring

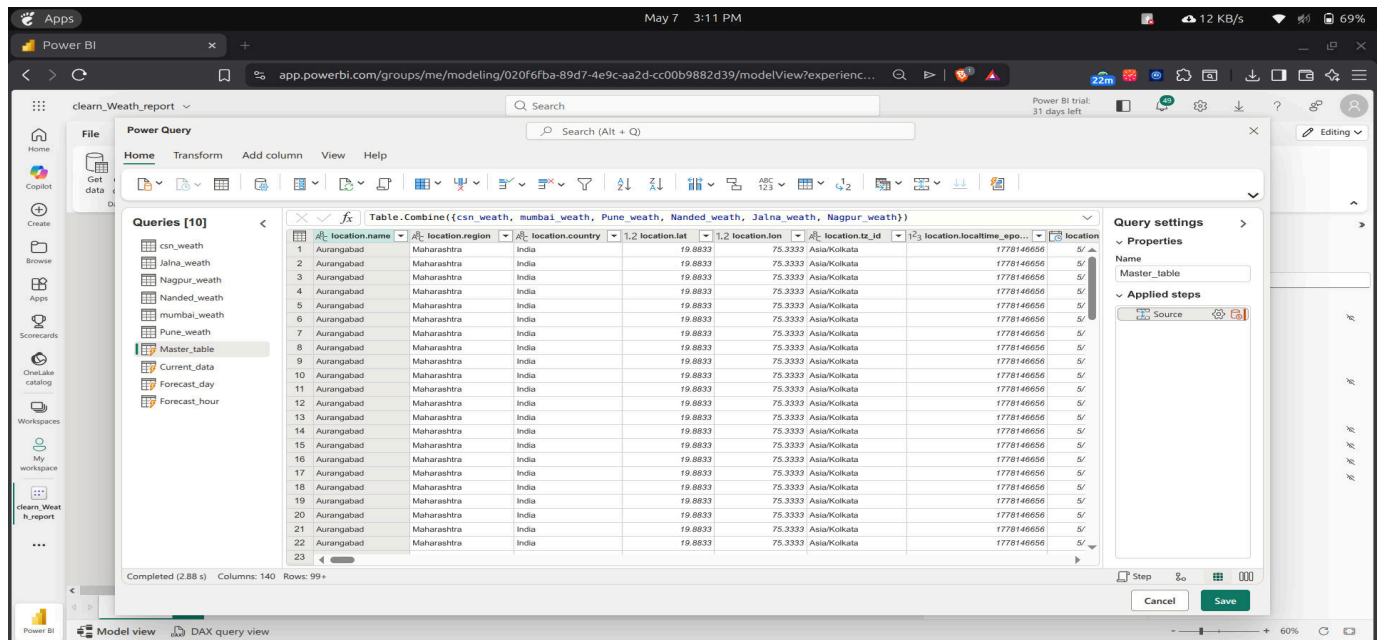


Figure 2.8 : Master Data Table

To optimize the data model and prevent data redundancy, the Master Data table was not used directly for visualization. Instead, three distinct, specialized tables were derived from it using Power Query:

- **Current Data Table:** Created by duplicating the Master Data, removing all forecast-related columns, and eliminating duplicated values to isolate the real-time weather metrics.

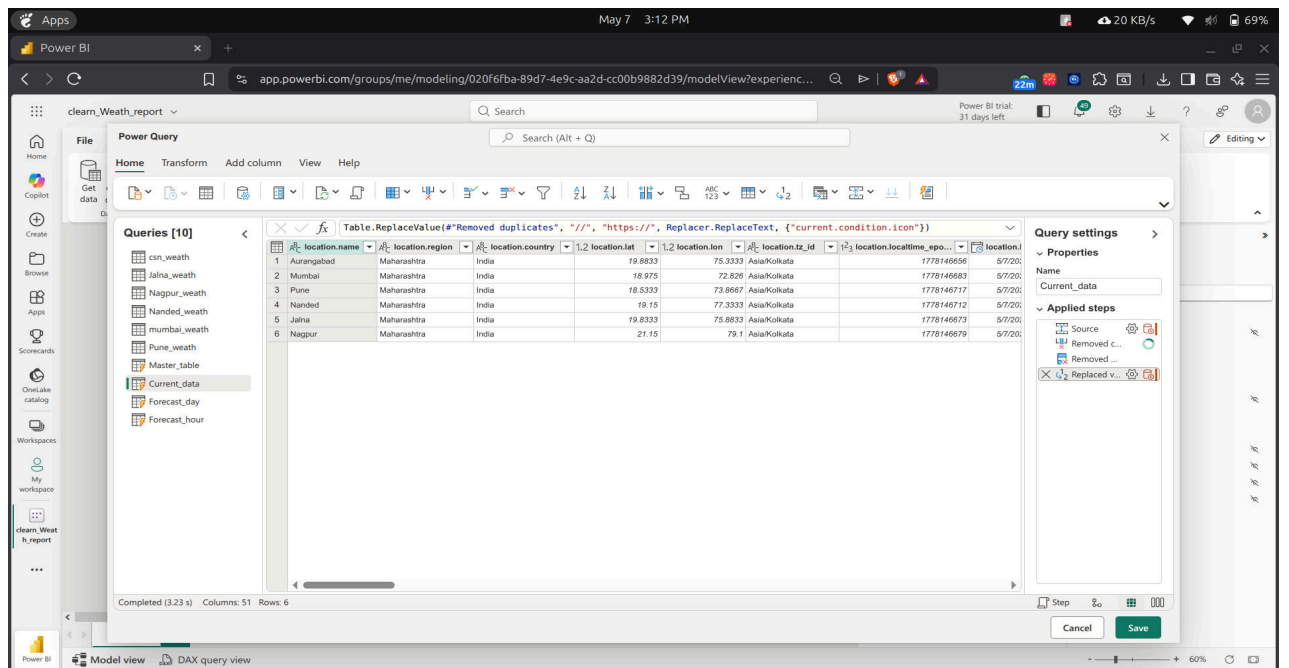


Figure 2.9 : Current Data Table

- Forecast Day Table: Created by duplicating the Master Data and stripping out both the 'current' metrics and 'forecast hour' columns, leaving a clean dataset dedicated to the daily forecast.

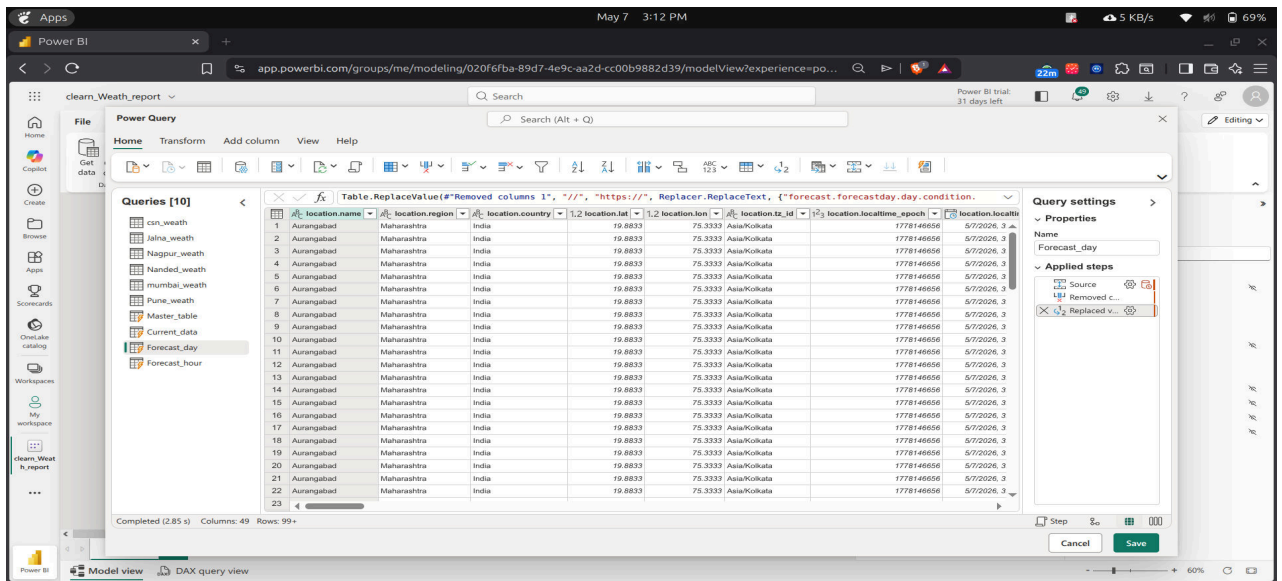


Figure 2.10 : Forecast Day Table

- Forecast Hour Table: Created by duplicating the Master Data and removing 'current' and 'forecast day' columns to focus strictly on granular, hourly predictions.

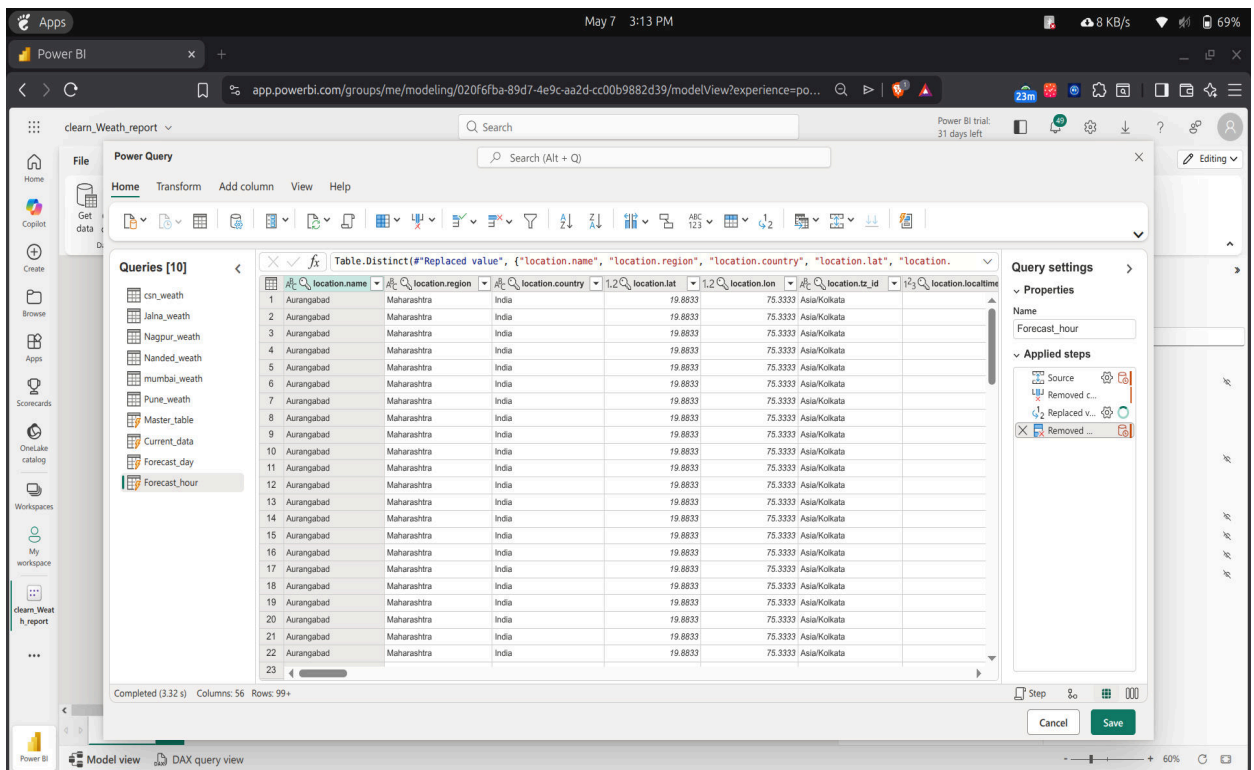


Figure 2.11 : Forecast Hour Table

2.2.3 Feature Engineering

New features were created within Power Query to enhance the dashboard's analytical capabilities:

- Day of Week Extraction: Extracted specific day names (Wednesday, Thursday, etc.) from the forecast dates to create intuitive X-axes for line charts.
- Weather Condition Grouping: Categorized raw AQI numerical data into text-based health bands (e.g., "Good", "Moderate").

2.2.3 DAX (Data Analysis Expressions) Implementation

To support the visualizations, custom DAX measures were written. To maintain a clean and organized data model, a dedicated, standalone table named **Measures** was created explicitly to house these calculations. This table stores all 17 custom DAX metrics (such as Average Humidity, Max Forecasted Heat, and specific AQI pollutants), separating the analytical logic from the raw data tables.

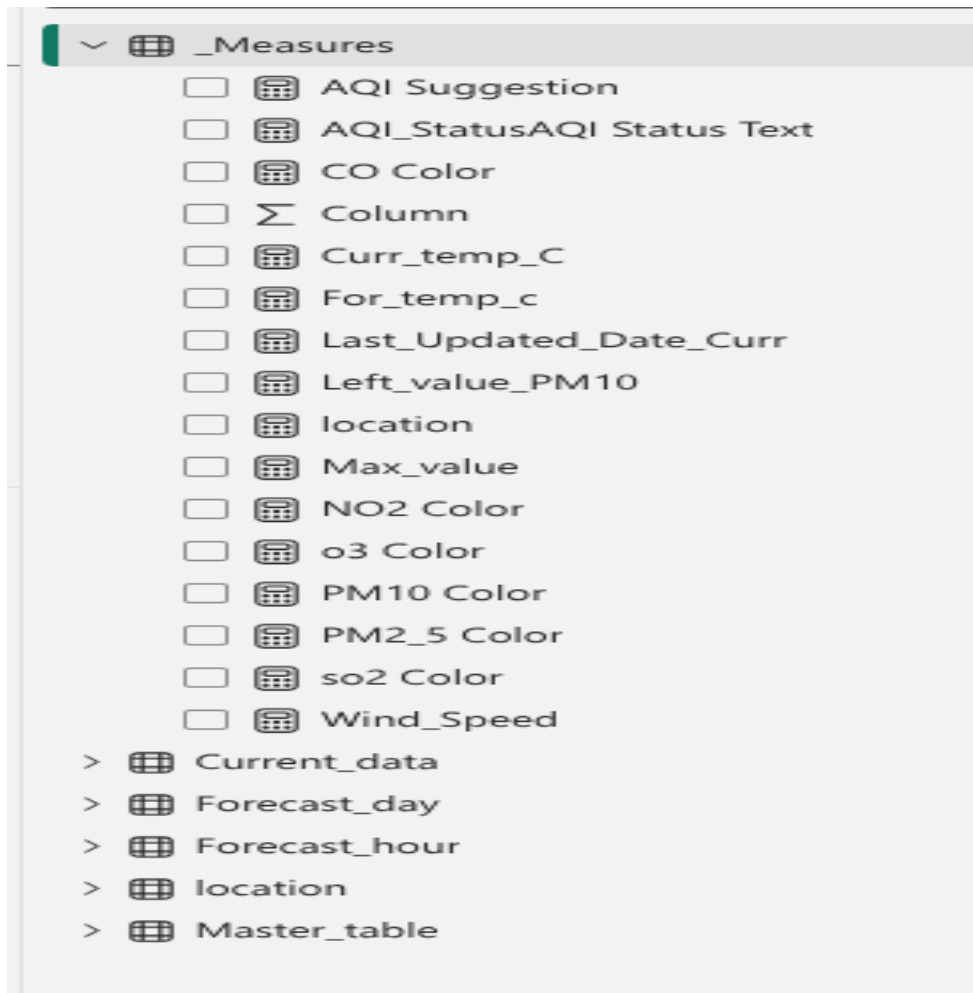


Figure 2.12: Measures Table

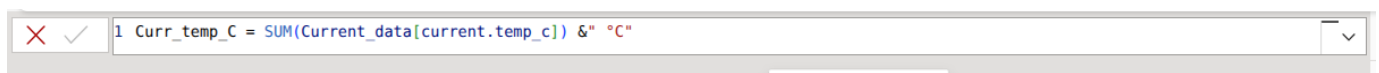


Figure 2.13: Current temperature DAX

```
1 For_temp_c = AVERAGE(Forecast_day[forecast.forecastday.day.avgtemp_c]) & " °C"
```

Figure 2.14: Forecast temperature DAX

```
1 AQI_StatusAQI Status Text =  
2 Var AQI = ROUND(SELECTEDVALUE(Current_data[current.air_quality.pm10]),0)  
3 RETURN  
4 SWITCH(  
5     TRUE(),  
6     AQI <= 50, "Good",  
7     AQI <= 100, "Moderate",  
8     AQI <= 150, "Unhealthy For Sensitivity",  
9     AQI <= 200, "Unhealthy",  
10    AQI <= 300, "Very Unhealthy",  
11    "Hazardous"  
12 )
```

Figure 2.15: AQI_status DAX

```
1 AQI Suggestion =  
2 Var AQI = ROUND(SELECTEDVALUE(Current_data[current.air_quality.pm10]),0)  
3 RETURN  
4 SWITCH(  
5     TRUE(),  
6     AQI <= 50, "Air is Clean and Healthy",  
7     AQI <= 100, "Acceptable air Quality, stay active",  
8     AQI <= 150, "Sensitive groups should reduce outdoor time ",  
9     AQI <= 200, "Limit Prolonged outdoor exertion",  
10    AQI <= 300, "Avoid outdoor activity if possible",  
11    "Stay indoors ,wear mask if outside !"  
12 )
```

Figure 2.16 : AQI suggestions DAX

```
1 PM2_5 Color =  
2 Var AQI = ROUND(SELECTEDVALUE(Current_data[current.air_quality.pm2_5]),0)  
3 RETURN  
4 SWITCH(  
5     TRUE(),  
6     AQI <= 50, "#43d946",  
7     AQI <= 100, "#fff570",  
8     AQI <= 150, "#ff9800",  
9     AQI <= 200, "#d99343",  
10    AQI <= 300, "#ff5b0f",  
11    "#d95243"  
12 )
```

Figure 2.17 : PM_2.5 DAX

```
1 CO Color =  
2 Var AQI = ROUND(SELECTEDVALUE(Current_data[current.air_quality.co]),0)  
3 RETURN  
4 SWITCH(  
5     TRUE(),  
6     AQI <= 50, "#43d946",  
7     AQI <= 100, "#fff570",  
8     AQI <= 150, "#ff9800",  
9     AQI <= 200, "#d99343",  
10    AQI <= 300, "#ff5b0f",  
11    "#d95243"  
12 )
```

Figure 2.18 : CO (carbon oxide) DAX

Once the transformed API data was loaded, a robust semantic model was established in Power BI's Model View. A structured star-schema approach was utilized to connect the central weather "Fact" table (containing the numerical metrics like temperature, humidity, and AQI) with corresponding dimensional tables (such as City Reference, Forecast Dates, and Weather Condition bands).

As shown in the schematic diagrams above, 1-to-Many relationships were established with bi-directional cross-filtering enabled where necessary. This relational structure ensures that when a user interacts with the dashboard—such as filtering by a specific city like "Nanded"—all associated visualizations across the page update synchronously and accurately without breaking the calculations.

2.3.3 Visualization Creation

Visualizations were crafted in the Report View. Each visual was selected based on established data visualization best practices to answer specific meteorological questions:

- **Line Charts:** Used to map the 7-day temperature trajectory (e.g., 34.9°C to 38.5°C), providing a clear visual curve of warming or cooling trends.
- **KPI Cards:** Utilized for instantaneous environmental readings (Humidity at 16%, Wind Speed at 12 Kph) requiring immediate user attention.
- **Gauge/Donut Charts:** Used to represent the Air Quality Index (e.g., PM10 at 48) and the percentage chance of precipitation.

2.3.4 DAX (Data Analysis Expressions) Implementation

To support the visualizations, custom DAX measures were written. These measures calculate dynamic values such as the Average Weekly Temperature, Maximum Forecasted Heat, and dynamic titles that change based on the user's city selection.

2.3.5 Filters and Interactivity

Slicers were integrated to enable dynamic data exploration. A dedicated geographic slicer allows the user to switch seamlessly between Nanded, Aurangabad, Jalna, and Mumbai. All visuals are set to cross-filter, meaning a selection in one chart dynamically updates the context of the others.

2.3.6 Dashboard Design Principles

The overall dashboard design was guided by principles of spatial organization and human-computer interaction:

- **Proximity & Grouping:** Related visuals (like all AQI metrics) are grouped together to reduce cognitive load.
- **Contrast & Hierarchy:** A dark-mode aesthetic was chosen. High-contrast colors (like orange for temperature lines and green for good AQI) were used to highlight critical information against the dark background.
- **F-Pattern Layout:** Key location data and current temperatures are placed at the top-left, leading the user's eye naturally across the 7-day forecast and down into specific metrics like wind and pressure.

Chapter 3: Implementation and Dashboard Explanation

This chapter provides a detailed walkthrough of the Power BI dashboard implementation, covering the interface overview, the specific visualization types used to represent meteorological data, the dashboard layout design, and the interactivity features that allow users to explore different cities.

3.1 Power BI Interface Overview

Power BI Desktop provides a multi-pane interface that supports the complete dashboard development process. It consists of three main views:

- **Report View:** Used to design the canvas, arrange visualizations, apply dark-mode themes, and configure interactive slicers.
- **Data View:** : Displays the flattened API JSON tables, allowing for validation of data types like fixed decimals for temperatures and DateTime formats for timestamps.
- **Model View:** Displays the star-schema relationships between the central weather fact table and the location dimensions.



Figure 3.1: Main Dashboard View showing the 7-Day Weather Forecast for Nanded

3.2 Key Visualizations Used

Visualizations were chosen based on their ability to clearly communicate specific environmental metrics.

3.2.1 Line and Area Charts (Trend Analysis)

Line charts are the most effective way to display time-series data. In this dashboard, a central line chart maps the 7-day temperature trajectory, tracking daily highs from Wednesday through Tuesday. This allows users to instantly visualize warming or cooling trends across the week, highlighting peak forecast temperatures such as 38.5°C.

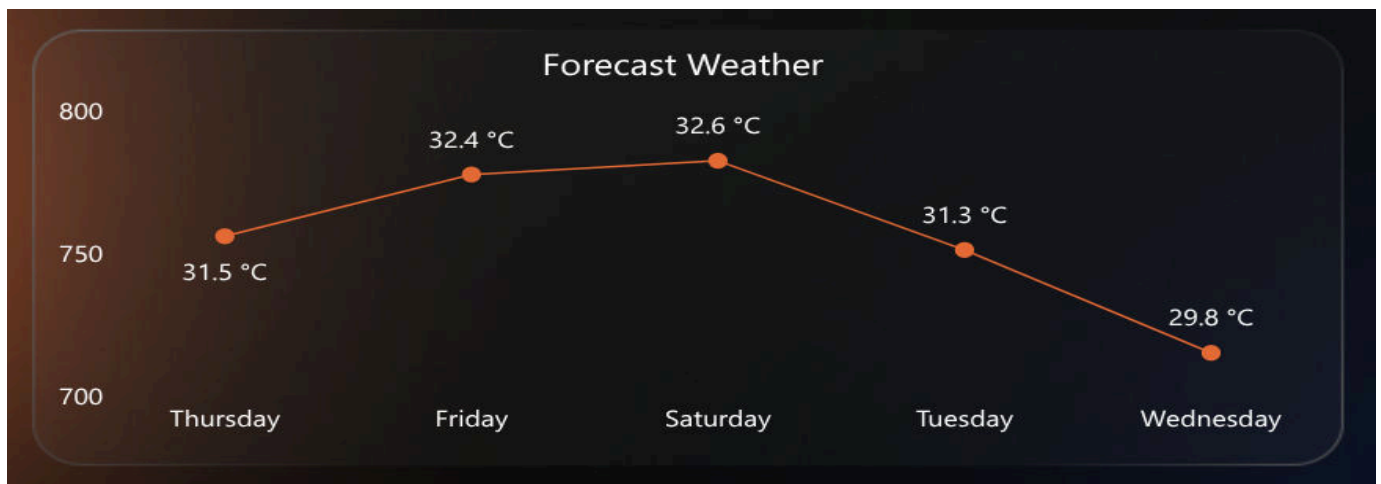
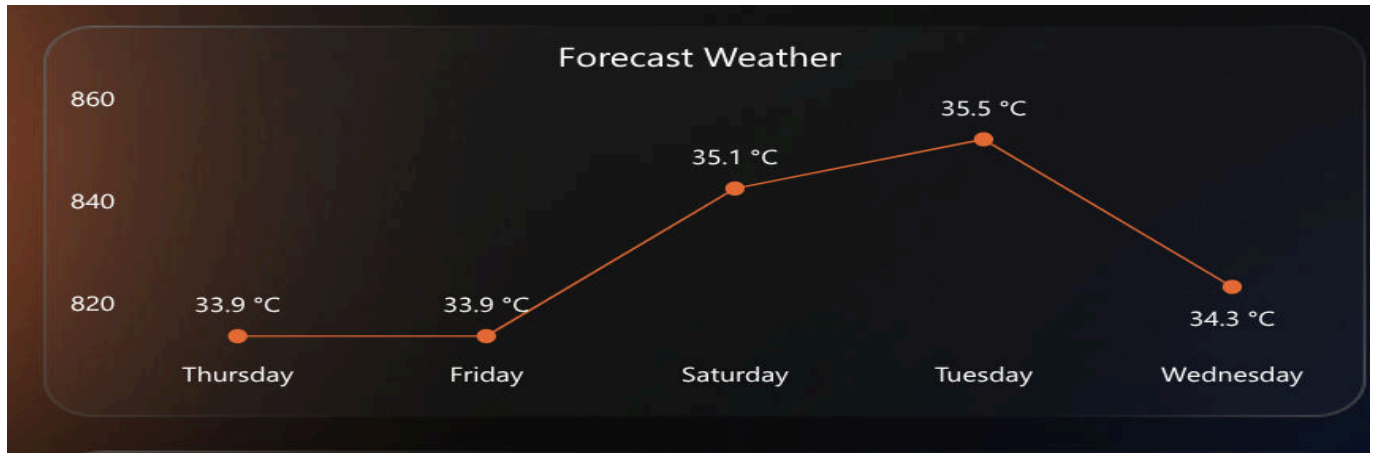


Figure 3.2 : 7-Day temperature trajectory comparison demonstrating localized forecast variations between Chhatrapati Sambhaji Nagar (top) and Pune (bottom).

3.2.2 KPI Cards (Current Metrics)

KPI (Key Performance Indicator) Cards display crucial, immediate metrics at a glance. They were deployed to show current environmental conditions that do not require historical trending:

- Humidity: Displayed as a percentage (e.g., 16%).
- Wind Speed: Displayed in Kph (e.g., 12 Kph).
- Atmospheric Pressure: Measured in inHg (e.g., 29.69).
- Visibility: Displayed in miles or kilometers (e.g., 10).

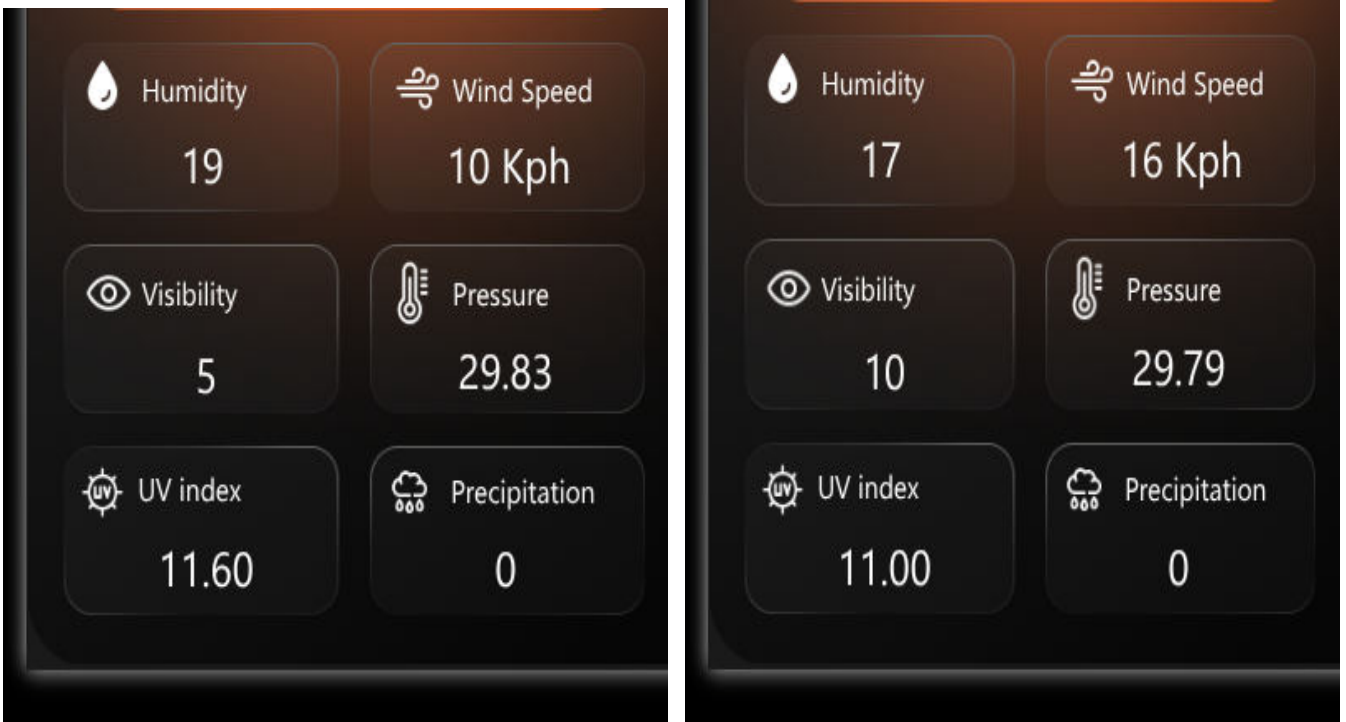


Figure 3.3: Dynamic KPI cards displaying real-time environmental metrics, updating automatically based on regional selection.

3.2.3 Matrix and Table Visuals (Detailed Breakdown)

A custom list/matrix visual was used on the left side of the dashboard to provide a clean, day-by-day textual breakdown of temperatures (e.g., Wednesday 35.9°C, Thursday 35.5°C). This provides precise numerical data for users who need exact figures rather than just visual trends.

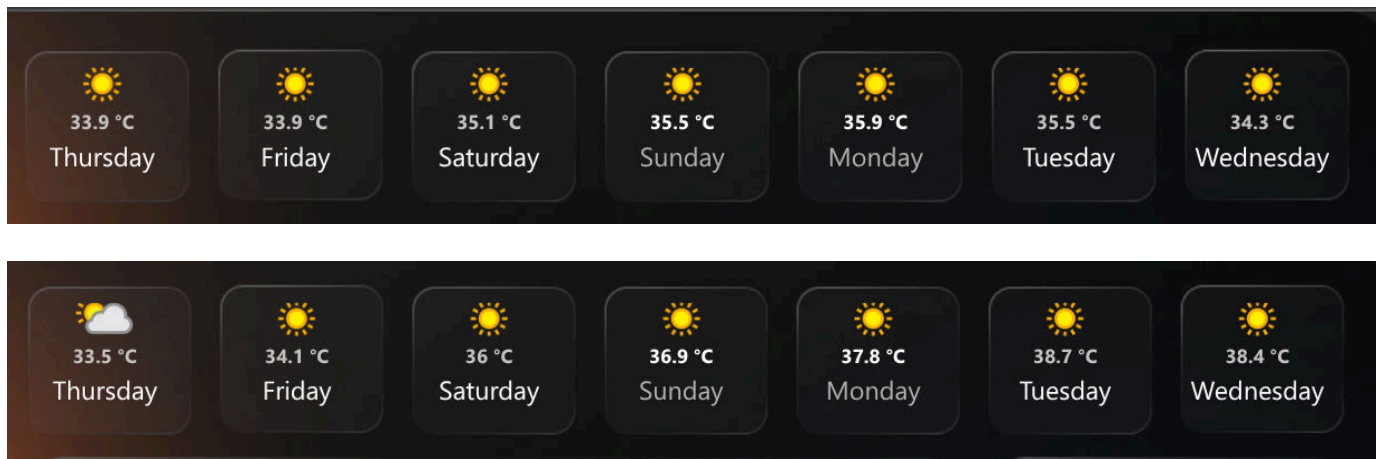


Figure 3.4: Day-by-day tabular breakdown demonstrating localized temperature forecasts for Chhatrapati Sambhajinagar (top) and Nanded (bottom).

3.2.4 Gauge and Donut Charts (Proportional Data)

- Air Quality Index (AQI): Donut charts effectively break down the components of the AQI, displaying specific pollutant levels like PM10 (e.g., 48) and O3 (e.g., 153) alongside a categorical health rating (e.g., "Good").
- Precipitation: Gauge or percentage visuals denote the probability of rain across the forecast days.



Figure 3.5: Air Quality Index (AQI) breakdown comparing specific pollutant levels (PM10, O3) across Chhatrapati Sambhajinagar (top) and Nanded (bottom).

3.3 Dashboard Layout Explanation

The dashboard interface was engineered as a consolidated, single-page application to reduce cognitive load. The layout uses spatial organization to group related meteorological concepts together into distinct visual zones.

3.3.1 Header and Identification (Top Section)

The topmost section serves as the primary context setter for the user.

- It dynamically displays the currently selected geographic region (e.g., Nanded, Mumbai).
- It includes a "Last Updated" timestamp to verify the freshness of the API data pull.
- It cleanly integrates astronomical data, showing Sunrise (e.g., 05:47 AM) and Sunset (e.g., 06:45 PM) times to provide daylight context.

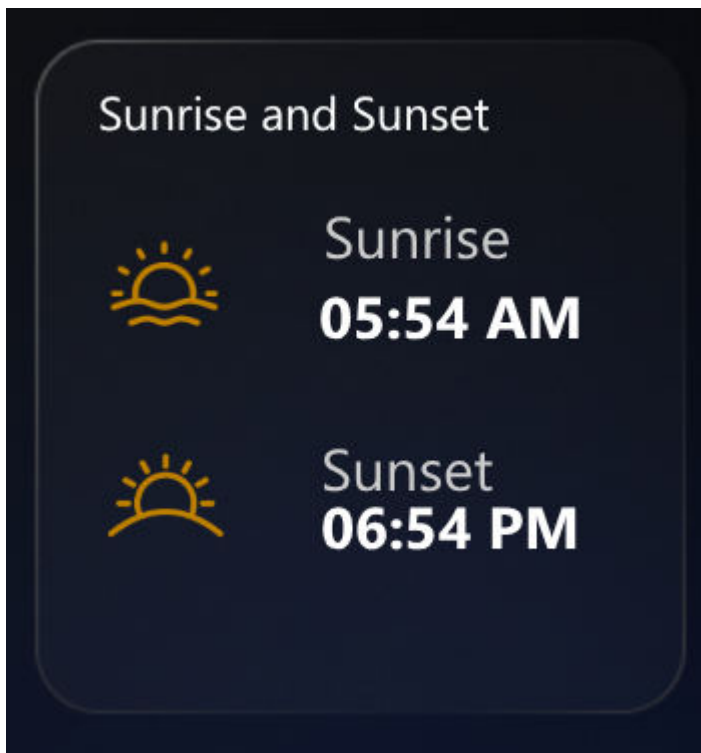


Figure 3.6: Astronomical context widget displaying dynamically fetched Sunrise and Sunset times.

3.3.2 Detailed Day-by-Day Panel (Left Section)

The left vertical quadrant acts as a data table for users requiring specific numerical values rather than visual trends.

- It lists the exact forecasted temperatures for each day of the week (e.g., Wednesday 35.9°C, Thursday 35.5°C).
- Weather condition icons (e.g., "Clear") are placed alongside the temperatures for quick visual recognition.

3.3.3 Primary Trend Visualization (Center Section)

The largest and most central portion of the dashboard canvas is dedicated to the "Forecast Weather" line chart.

- This placement draws the user's eye immediately to the most critical information: the 7-day temperature trajectory.
- The X-axis represents the days of the week, while the Y-axis scales dynamically based on the temperature range.

3.3.4 Secondary Environmental KPIs (Bottom Section)

The lower quadrant houses secondary, yet vital, environmental metrics. Grouping these prevents the main forecasting chart from becoming cluttered.

- Contains specific KPI cards for Humidity, Wind Speed, Visibility, and Atmospheric Pressure.
- Includes the Air Quality Index (AQI) donut chart, separating pollutant data (PM10, O3) from the main weather forecast.



Figure 3.7: KPIs

3.4 Filters and Interactivity

The dashboard moves beyond static reporting by incorporating dynamic user interactions, allowing for customized meteorological analysis.

3.4.1 Geographic Slicers (Location Filtering)

Slicers were implemented as the primary interactive control mechanism.

- Users can seamlessly switch context between various tracked cities (Nanded, Aurangabad, Jalna, Mumbai).
- Selecting a new city triggers a cascading DAX evaluation, instantly refreshing the entire dashboard to reflect the localized API data for that specific region.



Figure 3.8: Button Slicer

3.4.2 Visual Cross-Filtering

Power BI's native cross-filtering capabilities were enabled across the semantic model.

- If a user clicks on a specific data point within the center Forecast Weather chart (e.g., the 38.5°C peak on Tuesday), the surrounding KPI cards dynamically filter to isolate the specific atmospheric conditions anticipated for that exact day.

3.4.3 Dynamic Tooltips

Custom tooltips were configured for the visualizations. When a user hovers their cursor over specific elements—such as a node on the line chart or a segment of the AQI donut chart—a pop-up window reveals deeper, granular data points that are hidden from the main view to preserve visual cleanliness.

Chapter 4: Results and Analysis

This chapter presents the key insights derived from the interactive Power BI dashboard, focusing on regional temperature trends, atmospheric conditions, and the practical utility of the visualization.

4.1 Detailed Analytical Insights

4.1.1 Regional Temperature Variations

The implementation of the dashboard successfully highlighted significant geographical climate disparities within Maharashtra.

- Inland cities demonstrated higher thermal retention, with Nanded's 7-day forecast projecting a peak temperature of 38.5°C on Tuesday.
- Conversely, coastal and geographically distinct regions displayed milder conditions, with Mumbai recording a significantly cooler 30.1°C.
- The line chart visualizations effectively mapped these fluctuations, showing a slight cooling dip before the peak heat wave (e.g., dropping to 34.9°C on Friday before rising again).

4.1.2 Atmospheric and Health Metrics Analysis

Beyond basic temperature tracking, the dashboard provided critical environmental health data:

- Air Quality Index (AQI): The integration of specific pollutants like PM10 (recording at 48) and O3 (recording at 153) allowed for an overall categorical assessment of air quality, maintaining a "Good" rating for the observed period.
- Dry Conditions: Low humidity levels (16%) paired with moderate wind speeds (12 Kph) and zero predicted precipitation (100% chance of not raining) indicated highly dry regional conditions.

4.2 Before vs After Visualization

- Before (Raw API Data): The initial data state consisted of nested, highly technical JSON payloads. Timestamps were in Unix format, and meteorological metrics were buried in arrays, making it impossible for a non-technical user to extract immediate meaning or forecast trends.
- After (Power BI Dashboard): The raw JSON was transformed into a clean, graphical interface. Users can now instantly comprehend a full week's forecast through visual trend lines, gauge charts, and easily readable KPI cards without writing any code or parsing data arrays.

4.3 Decision-Making Insights

The transformation of this data supports several key decision-making processes:

- Agricultural Planning: Farmers can use the 7-day temperature and precipitation forecasts to optimize irrigation schedules, especially during dry spells with 16% humidity.
- Public Health: Real-time AQI tracking allows vulnerable populations to make informed decisions about outdoor activities based on PM10 and O3 levels.
- Travel and Logistics: Visibility metrics (e.g., 10 miles) and wind speed data assist in localized transit and logistical planning.

Chapter 5: Conclusion

5.1 Summary of Findings

This project successfully developed an automated, interactive dashboard using Microsoft Power BI to analyze real-time weather conditions across multiple cities. By migrating away from static CSV files, the project established a functional mini-ETL pipeline using Power Query to ingest, clean, and model live JSON data from a RESTful Weather API. The resulting dashboard provides clear, dynamic insights into temperature trends, air quality, and atmospheric pressure, proving that complex environmental data can be effectively simplified for end-user analysis.

5.2 Limitations of the Project

- **Lack of Historical Data:** Because the API fetches a rolling 7-day forecast and current conditions, the dashboard currently lacks the ability to perform year-over-year historical climate comparisons.
- **API Constraints:** The frequency of data updates is limited by the specific rate limits and call allowances of the third-party Weather API.
- **Internet Dependency:** Unlike static datasets, the dashboard requires an active internet connection to refresh the web queries and update the visualizations.

5.3 Future Scope and Enhancements

- **Automated Alerting (Power Automate):** As demonstrated in the proposed architecture above, Microsoft Power Automate could be integrated to trigger automated email or SMS alerts when the API detects severe weather conditions or dangerous AQI spikes.
- **Data Warehousing:** Implement a cloud SQL database (such as Azure SQL) to store the daily API pulls. Over time, this would build a proprietary historical dataset, enabling advanced machine learning predictions and seasonal trend analysis.
- **Expanded Geographic Scope:** Scale the semantic model to include a wider range of states and national territories, mapping broader climatic shifts across India.

References

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Appendix

Appendix A: Key DAX Measures Reference

Measure Name	DAX Formula
AQI Suggestion	<code>AQI Suggestion = SWITCH(TRUE(), [pm10] <= 50, "Air is Clean and Healthy", [pm10] <= 100, "Moderate Air Quality", "Poor Air Quality")</code>
AQI_Status	<code>AQI_Status = IF([pm10] <= 50, 1, IF([pm10] <= 100, 2, 3))</code>
AQI Status Text	<code>AQI Status Text = SWITCH(TRUE(), [pm10] <= 50, "Good", [pm10] <= 100, "Moderate", "Unhealthy")</code>
Chance of rain	<code>Chance of rain = AVERAGE('Forecast_day'[day_daily_chance_of_rain]) / 100</code>
co	<code>co = MAX('Current_data'[co])</code>
CO Color	<code>CO Color = IF([co] <= 4.4, "#00FF00", "#FF0000")</code>
ConditionText	<code>ConditionText = MAX('Current_data'[condition_text])</code>
ConditionText_forecast	<code>ConditionText_forecast = MAX('Forecast_day'[day_condition_text])</code>

Curr_temp_C	Curr_temp_C = MAX('Current_data'[temp_c])
For_temp_c	For_temp_c = MAX('Forecast_day'[maxtemp_c])
gb-defra-index	gb-defra-index = MAX('Current_data'[gb-defra-index])
Humidity	Humidity = AVERAGE('Current_data'[humidity])
Last_Updated_Date_Curr	Last_Updated_Date_Curr = MAX('Current_data'[last_updated])
LocationName	LocationName = MAX('Location'[name])
MaxTemp	MaxTemp = MAX('Forecast_day'[day_maxtemp_c])
MinTemp	MinTemp = MIN('Forecast_day'[day_mintemp_c])
no2	no2 = MAX('Current_data'[no2])
o3	o3 = MAX('Current_data'[o3])
pm10	pm10 = MAX('Current_data'[pm10])

pm2_5	pm2_5 = MAX('Current_data'[pm2_5])
Pressure	Pressure = MAX('Current_data'[pressure_in])
so2	so2 = MAX('Current_data'[so2])
Visibility	Visibility = MAX('Current_data'[vis_miles])
WindSpeed	WindSpeed = AVERAGE('Current_data'[wind_kph])

Appendix B: Data Cleaning Checklist (Power Query)

Step	Action Taken
1. API Connection & JSON Parsing	Established web connections to the RESTful Weather API for 6 cities and parsed the raw nested JSON payloads.
2. Master Data Consolidation	Appended and combined the parsed JSON queries for all 6 cities into one central Master Data dataset.
3. 'Current Data' Derivation	Duplicated Master Data, removed all forecast-related columns, and removed duplicated values to create a clean Current Data table.
4. 'Forecast Day' Derivation	Duplicated Master Data, and removed all 'current' and 'forecast hour' columns to create the Forecast Day table.

5. 'Forecast Hour' Derivation	Duplicated Master Data, and removed all 'current' and 'forecast day' columns to create the Forecast Hour table.
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Appendix C: Dashboard Technical Specifications

Specification	Detail
Tool Used	Microsoft Power BI Desktop (Latest Version)
Data Source Type	RESTful Web API (JSON Format via WeatherAPI.com)
Data Scope	6 Cities in Maharashtra, 7-Day Rolling Forecast & Current Conditions
Refresh Methodology	Dynamic Web Query (API Polling) via Power Query
Number of Dashboard Pages	1 Consolidated Interactive Canvas
Primary Visualizations	Line Chart (Trend), Matrix (Daily Breakdown), KPI Cards, Donut Chart (AQI)
Interactive Slicers	1 Geographic Slicer (Dynamic City Selection)